

ALEXANDRE LEUCKERT KLEIN

Senior Software Engineer | Full-Stack, Embedded & IoT Systems | Technical Leadership | Startup & Deep Tech

+5551982273320 contact@alexandre.dev alexandre.dev Novo Hamburgo, Brazil



FIND ME ONLINE

GitHub
github.com/AlexandreLKlein

LinkedIn
linkedin.com/in/alexandre-l-klein

SUMMARY

Senior Software Engineer with experience building end-to-end products across embedded systems, backend platforms, and customer-facing applications. I've helped startups turn ideas into production-ready systems by working across the entire technology stack—from electronics and firmware to cloud services and modern web applications. My work has directly contributed to the development of core products and supported the successful raise of multiple millions of dollars in investment.

My experience includes:

- High-performance real-time processing of massive point cloud datasets
- Embedded firmware and hardware development (ARM, bare-metal, RTOS)
- VR applications and Unreal Engine plugins in C++
- ERP, CRM, and POS platforms integrating payments, fiscal compliance, IoT, and business operations using Laravel and PHP
- Technical leadership, architecture, and cross-functional product development
- Foundations in offensive security and cyber intelligence

I enjoy solving complex problems, taking ownership of critical systems end-to-end, and building products that bridge hardware, software, and real-world business needs.

EXPERIENCE

Firmware & Hardware Engineer

Alignment Products Inc

03/2024 - Present USA

- Designed and implemented firmware for the company's flagship product using C
- Developed hardware architectures and integrated embedded systems components
- Participated in product validation and technical decision-making
- Worked on low-level software interacting directly with hardware peripherals
- Contributed to the successful development of a commercial product supporting investment fundraising efforts
- Collaborated across disciplines in a fast-paced startup environment
- Technologies: C, ARM, RTOS, Firmware Development, Embedded Systems, Hardware Design

Technical Lead / Senior C++ Software Engineer

Miris / Illusio Inc

08/2021 - 03/2024 USA

- Architected and developed core components for a high-performance platform capable of real-time processing of billions of 3D points, utilizing C++17/20.
- Led technical initiatives, mentored team members, and contributed to strategic architectural decisions facilitating project success.
- Optimized application performance and throughput in critical workloads through profiling and architectural improvements.
- Built networking and streaming functionality, integrating TCP/IP, Sockets, and Boost.Asio to support real-time data processing.
- Delivered a key product that enabled multi-million-dollar fundraising, showcasing technological innovation and market potential.
- Developed integrations for VR environments and plugins for Unreal Engine, expanding platform capabilities.
- Implemented multithreaded and asynchronous components for enhanced performance using modern C++ technologies.

Software Developer

FishTV

09/2018 - 08/2021 Brazil

- Developed and maintained ERP and CRM systems supporting company operations
- Built software platforms used for championship and event management
- Designed and implemented REST APIs and backend services
- Developed production-ready software across the full development lifecycle
- Enhanced operational efficiency and contributed to business growth through technology initiatives
- Technologies: C#, Laravel, JavaScript, REST APIs, SQL, Git

KEY ACHIEVEMENTS

ERP, CRM & POS for Breweries and Pubs
Increased revenue 12.5x through the development of an integrated platform.

Real-Time 3D Processing at Scale
Built high-performance systems for real-time processing of billions of 3D points, demonstrating deep expertise in point cloud performance, streaming, and large-scale visualization.

Product Contribution Supporting Fundraising
Contributed to core product development that helped support more than \$5 million in fundraising, linking technical execution to business outcomes.

Technical Leadership Across Complex Systems
Led development across cross-platform, embedded, and backend systems, helping improve production efficiency and technical execution in fast-moving environments.

PATENTS

Systems and methods for augmented reality viewing based on directly mapped point cloud overlays — Active — US 11657586
US patent in augmented reality and directly mapped point cloud overlays. 05/2023

Systems and methods for compressing three-dimensional image data — Active — US 11710258
US patent in three-dimensional image data compression. 07/2025

Systems and methods for augmented reality viewing based on directly mapped point cloud overlays — Active — US 11538228
US patent in augmented reality and directly mapped point cloud overlays. 12/2022

Systems and methods for decompressing three-dimensional image data — Active — US 11893760
US patent in three-dimensional image data decompression. 02/2024

Systems and methods for efficient rendering and processing of point clouds using textures — Active — US 11615578
US patent in efficient point cloud rendering and processing using textures. 03/2023

EXPERIENCE

Lead Software Developer

Klein Sistemas Eletrônicos

01/2007 - 01/2018 Location

- Started in technical roles and progressed to Lead Software Developer, taking ownership of end-to-end delivery across hardware and software projects.
- Developed embedded and application software in PHP, Laravel, C++, and C# for products spanning internal tools and customer-facing systems.
- Contributed to hardware projects including circuit design and system integration, bridging electronics and software requirements.
- Collaborated on technical decisions, implementation planning, and delivery across multidisciplinary projects in a growing environment.

SKILLS

Languages

PHP, Laravel, C++, C#, C, JavaScript, TypeScript

Build & DevOps

CMake, Git, CI/CD, Bash, PowerShell

Security

Secure software practices, Cybersecurity fundamentals, Offensive Security concepts, Endpoint security interest, Threat awareness

Cloud

Azure, GCP and AWS concepts, Scalable architectures, Distributed systems

Embedded Systems

Hardware design, Circuit design, Embedded systems, Firmware development, ARM microcontrollers, RTOS, Hardware integration, Electronic architecture design

Networking

TCP/IP, UDP, Sockets, HTTP/HTTPS, REST APIs, Boost.Asio, Client-server architectures, Streaming systems, OpenSSL, mbedTLS, Certificate-based communication

PROJECTS

ERP, CRM & POS for Breweries and Pubs

03/2024 - Present Novo Hamburgo, Brazil

Developed an integrated ERP, CRM, and POS platform for breweries, supporting operations, inventory, loyalty, production, reporting, fiscal compliance, payments, and IoT brewing. Built with Laravel, Vue.js, and Vuetify; central for daily operations and management.

- Enabled a 12.5× increase in gross revenue while substantially reducing operational losses and waste through the implementation of an integrated ERP, CRM, and POS platform.
- Significantly reduced inventory losses and waste through real-time stock control, production tracking, and automated consumption reconciliation.
- Transformed brewery and pub operations by replacing fragmented manual processes with an integrated ERP, CRM, and POS platform that became the single source of truth for the business.
- Created a loyalty ecosystem converting visitors into repeat customers, enhancing relationships and revenue.

EDUCATION

Associate Degree in Brewing Technology

Universidade Pitágoras Unopar Anhanguera

01/2022 - 07/2023 Location

Postgraduate Degree in Offensive Security and Cyber Intelligence

Universidade Feevale

01/2019 - 01/2020 Location

Bachelor of Science in Computer Science

Universidade Feevale

2005 - 2011 Novo Hamburgo, Brazil